Reproducibility Report: Kirkland and Coppock (2018)

Candidate Choice Without Party Labels: New Insights from Conjoint Survey Experiments

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This repository holds the actively maintained replication code for Kirkland and Coppock (2018), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and Coppock

(2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1007/s11109-017-9414-8
Replication archive	10.7910/DVN/WSUHI3
Pre-analysis plans	none

The data are not redistributed here. The deposit lives at Harvard Dataverse and that is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` records the Dataverse file identifiers, the UNF of each ingested data file, and two checksums per file: the MD5 of the bytes Dataverse serves, which is what this code was written against, and the MD5 Dataverse publishes. Here the two agree for all 10 files. They do not always, so the script verifies against the served bytes and reports any disagreement. Either way the exact bytes are pinned in version control even though the bytes themselves are not.

`recovered/` is different and is committed. The observational analysis behind Table 1 depends on a data file and a script that the deposit does not contain, though its own README lists both. The authors supplied them, and since they exist nowhere else this repository carries them.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it, starting from `published_claims.csv`, an extraction of every numeric token in the article and the online appendix. `maintained/in_text_claims.R` prints each of those numbers beside the sentence that states it. `original/` is created by the download script and is deliberately absent from the repository. `errata.qmd` renders `kirkland_coppock_2018_errata.pdf`, a short note listing the article's misstated numbers. This README is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains, so nothing in the chain is more restrictive than the archive itself. See `LICENSE`.

To reproduce. Clone or download the repository, open `kirkland_coppock_2018.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposited archive from Dataverse, verifies its 10 checksums, produces every published table and figure into `maintained/output/`, rebuilds the ground truth table from those outputs, prints every published number beside the sentence that states it, and verifies the deposit again. It takes about a minute. Individual scripts can be run on their own in any order;

only `ground_truth/build_ground_truth.R` and `maintained/in_text_claims.R` depend on the others having run first.

Required packages: tidyverse, estimatr, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory and the scripts work equally well under `Rscript` outside RStudio. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check**, and the CSV and PNG output should come back byte-identical. The eleven PDF figures always show as changed, because a PDF records the time it was written; compare their PNG twins instead.

1 Paper Overview

Citation: Kirkland, P. A. and Coppock, A. (2018). Candidate choice without party labels: New insights from conjoint survey experiments. *Political Behavior*, 40(3), 571–591.

Research question: When a ballot carries no party label, what do voters use instead? The paper argues that without the partisan shortcut voters fall back on other indicators of fitness for office, political experience above all.

Design: Three pieces of evidence. An observational record of 1,010 US mayoral elections held between 1945 and 2007 across 237 municipalities, comparing the political experience of winners under partisan and nonpartisan ballots. Then two conjoint survey experiments running the same design on different samples: an MTurk convenience sample (N = 1,204) and a nationally representative YouGov sample (N = 1,200). Each subject judged five successive elections between two candidates whose race, gender, age, career experience, political experience and, in half the elections, party were fully randomized. Outcomes are candidate choice and a 0 to 100 competence rating. Estimation is OLS with standard errors clustered by respondent, and effects are compared across election types by interacting every attribute with an indicator for the nonpartisan ballot.

Main finding: Political experience carries substantially more weight when the party label is withheld. In the MTurk sample the political experience coefficients are about 10 percentage points larger in nonpartisan than in partisan elections, and the same interaction appears in the YouGov sample. The competence ratings move the same way, which is the paper's evidence for the mechanism it proposes. The two samples' estimates correlate at 0.95.

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2 Summary

Two questions, answered before the detail.

2.1 Does the deposited archive run?

Not as deposited, and the reasons are all about files rather than about code.

The YouGov data cannot be loaded under the name the scripts use. The deposit contains `yougov_replication.xlsx`. It is not a spreadsheet: it is an RData binary carrying the wrong extension, uploaded that way and served that way ever since. Both scripts that need it call `load("yougov_replication.rdata")`, so `yougov_replication.R` and `compare_samples_replication.R` fail on their first data line. Renaming the file, or loading it under its real name, fixes both, and the extension is the whole of the problem: the bytes are a valid RData file and `load()` reads them without complaint.

Three of the twelve files the archive's README promises were never deposited. The README lists `observational_mayors_replication.rdata`, `observational_replication.R` and `observational_codebook.txt`, along with CSV copies of all three data files. The Dataverse record has ten files and none of those six are among them. Table 1 and the paragraph of observational results that follows it therefore cannot be reproduced from the deposit at all. The authors supplied the missing data and script separately; they are in `recovered/` and the rewrite reads Table 1 from there.

Every data path in the archive is mis-cased. The scripts call `load("mturk_replication.rdata")` and `load("base_categories.rdata")`; the deposited files are `mturk_replication.RData` and `base_categories.RData`. macOS and Windows do not care. Linux does, which is where most people who clone a replication archive today are running it.

With the YouGov file renamed, all three deposited scripts run to completion in a current R and produce their figures. The only warning is `geom_errorbarh()`, deprecated in ggplot2 4.0.0. Nothing in the archive writes a file, so running it leaves the deposit untouched, and re-verifying `original/` against the manifest after a full run confirms all 10 files unchanged.

2.2 Does the maintained rewrite reproduce the paper?

Yes for everything the paper prints, with exceptions in the observational paragraph on page 5, in one sentence describing Figure 3, and in two sentences describing patterns in the heterogeneous-effects figures.

Table 1: Reproduction verdict by component

Component	Verdict
Table 1 (24 cells plus both totals)	Reproduces exactly
Table 1 chi-squared statistic	Reproduces (21.2)
Table 1 chi-squared p-value	Reproduces; the archive's test is an unseeded simulation and 0.002 is inside its range
Text p. 5, 1,010 elections and the 1945 to 2007 span	Reproduce exactly
Text p. 5, 225 municipalities	Does not reproduce (237 here)

Text p. 5, 779 elections	Reproduces exactly
Text p. 5, 62 and 61 per cent	Reproduce, but the two labels are transposed in the sentence
Text p. 5, p = 0.791	Does not reproduce (0.852 here)
Text p. 5, 63 and 48 per cent	Not verifiable; candidate occupation is not in the recovered data
Table A.1 party identification panel and Ns	Reproduces exactly
Table A.1 remaining panels	Not verifiable; the deposit carries no other respondent demographics
Figures 3 to 8, B.1, B.2, C.3, C.4	Reproduce; all 729 estimates and standard errors agree with the archive to 4e-13
Text on Figure 3, 25 to 30 percentage points	Range is tight at both ends; the estimates run 22.7 to 32.0
Text on Figure 3, about 10 percentage points	Reproduces (9.1 to 11.7)
Figures 5 and 6, Republicans prefer white candidates	Holds on YouGov and not on MTurk, where Republicans prefer Asian candidates by 6.1 points
Figure 7, Republicans show no competence gap by election type	Does not hold; the Republican estimate for Mayor excludes zero
Discussion, correlation of 0.95	Reproduces (0.9494) on the archive's specification

Every numeric token in the article and the online appendix was extracted by hand into `ground_truth/published_claims.csv`, 178 of them, and each classified. 100 are quantities the pipeline should produce, 41 are claims about shape, sign or ordering that need a computed truth value, and the rest are scale endpoints, category labels, and numbers copied from other papers. The ground truth carries 164 rows, one for every claim either instrument can reach; 112 of them can be evaluated and 104 come out right. Of the 52 that cannot, 45 are quantities the deposit does not record at all, 6 are descriptions whose wording sets no threshold to test against, and one is a sentence that names neither a sample nor an election type.

The article contradicts its own data on 8 claims. Six of them are misstated numbers in sentences, gathered into the 4 entries of `kirkland_coppock_2018_errata.pdf`. The other two are descriptions of the heterogeneous-effects figures that the estimates behind those figures do not support, and they are discussed under Errata below.

The p-value that does not reproduce. The paper reports that among the 779 elections in which the two candidates had different levels of political experience, the more experienced candidate won about 62 per cent of the time in partisan elections and 61 per cent in nonpartisan elections, and that "this difference is not statistically significant (p = 0.791)." The subset size reproduces exactly, and so do both percentages, though the sentence attaches them to the wrong ballot types: the data give 62.1 per cent nonpartisan and 61.3 per cent partisan. The p-value does not reproduce under any standard test of that comparison. A regression with HC2 errors gives 0.852, a chi-squared test gives 0.919, and Fisher's exact test, a logit and a t test all land near 0.85. No deposited or recovered script computes this quantity, so there is no code to inspect. The stated conclusion, that the difference is not significant, is unaffected either way: 0.791 and 0.85 lead to the same reading.

The chi-squared p-value is not a fixed number. The recovered script calls `chisq.test(simulate.p.value = TRUE)` and sets no seed, so its p-value is a Monte Carlo draw. Over 500 replications of that same call the p-value runs from 5×10^{-4} to 0.004, and 24 per cent of draws round to the published 0.002. The published value is a perfectly ordinary

draw from the archive's own procedure, not a discrepancy. The asymptotic p-value, which is exact, is 7.44×10^{-4}.

The correlation of 0.95 depends on an inconsistency inside the archive. `yougov_replication.R` weights the YouGov main-effects models with the survey weights. `compare_samples_replication.R`, which produces the correlation, refits the same four models without them. The rewrite keeps the archive's unweighted specification, which is what the published 0.95 requires: weighting those models instead gives 0.9184, which rounds to 0.92.

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3 Original Archive Reproducibility

Archive source: Harvard Dataverse, DOI 10.7910/DVN/WSUHI3, version 1.0, ten files.

Table 2: Original archive check (current R environment)

File	Status	Issue
mayors_source.R	Runs	Helper functions; sourced by every script
mturk_replication.R	Runs	load() path is mis-cased and would fail on a case-sensitive filesystem; geom_errorbarh() deprecated
yougov_replication.R	FAILS	load('yougov_replication.rdata'): the deposit contains yougov_replication.xlsx, an RData binary under the wrong extension
compare_samples_replication.R	FAILS	Same missing file name
observational_replication.R	ABSENT	Listed in the archive README; not deposited. Recovered from the authors
observational_mayors_replication.rdata	ABSENT	Listed in the archive README; not deposited. Recovered from the authors
observational_codebook.txt	ABSENT	Listed in the archive README; not deposited
CSV copies of the three datasets	ABSENT	Listed in the archive README; not deposited

The extension anomaly is visible at the Dataverse level, not just in the download. The dataset API reports `originalFileName` as `yougov_replication.xlsx` for that file, and Dataverse ingested it into a tabular representation named `yougov_replication.tab`. Whatever the upload looked like, the deposited bytes are an RData file and the name has never matched them. `download_original.R` writes the file under the deposited name and `maintained/helpers.R` loads it under that name, because renaming a deposited file inside a reproduction is how a defect stops being visible to the next reader.

What the archive computes but the appendix does not print. Each of the two conjoint scripts contains seven figure blocks. Four correspond to published figures. The remaining three per script build a valence-index analysis, a policy-index analysis and a satisficing analysis, and each script also assembles regression tables it labels D.3, D.5 and so on. The published online appendix has sections A, B and C, no section D, and no valence or policy figures. The

paper's Discussion cites "Appendix C.2" for the policy-goal results and "Appendix C.3" for the satisficing results; the appendix as published has neither of those subsections, and its satisficing figures are numbered B.1 and B.2. The online appendix is also dated June 2016, carries a different subtitle from the article, and refers to the main-text conjoint figures as Figures 1 and 2 where the article numbers them 3 and 4. It is an earlier draft of the appendix, published alongside a later draft of the paper.

Additional patterns (warnings or fragility, not errors):

- `geom_errorbarh()`, deprecated in ggplot2 4.0.0.
- `position_dodgev()` from `ggstance`, reached through `coefplot`, which the scripts install from GitHub with the comment "it appears that we need version > 1.2.4.9000."
- `gather` / `spread` throughout.
- `rm(list = ls())` at the top of every script.
- The difference-panel coefficients are selected by hardcoded row index (`fit_3_df[19:33, ]`, `[22:39, ]`, `[19:35, ]`, `[22:41, ]`, four different windows), and their labels are then overwritten by assignment from another model's row names. Adding a level to any attribute silently mislabels the entire panel.
- `cl()` recomputes the cluster vector from the unfiltered data, so wherever the outcome has missing values the script has to pass `filter(..., !is.na(comp))` by hand. It does, for the competence figures. Nothing enforces it.

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4 Errata

The rewrite corrects no analytical error in the deposited code, because none of the discrepancies above is one. What the comparison does turn up is misstated numbers in the article itself.

Four sentences carry a number the article's own data do not support. They are set out in `errata.qmd`, which renders to `kirkland_coppock_2018_errata.pdf` at the repository root: the transposition of the two ballot types in the experience-advantage comparison, the p-value of 0.791 that reproduces under no test, the count of municipalities in the mayoral dataset, and the range of political experience effects in the sentence describing Figure 3. Every corrected value in that note is computed at render time from `maintained/output/`, so it cannot go stale in the way the article did. None of the four changes a conclusion.

Two descriptions of the figures do not hold. These are not misstated numbers and are not in the errata note, since the correction is a rewording rather than a value.

- "Republican respondents marginally prefer white candidates while Democrats marginally prefer nonwhite candidates" holds in the YouGov sample, where all three Republican non-white estimates are negative. It does not hold on MTurk, where Republican respondents

prefer Asian candidates to white candidates by 6.1 percentage points while preferring white candidates to Black and Hispanic ones.

- "Democratic respondents (but not Republican respondents) rate such candidates as even more competent in the absence of party labels" reads the difference column of Figure 7 as null for Republicans. Two of the four Democratic intervals exclude zero, and so does one of the four Republican ones, for candidates who have been mayor.

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5 Ground Truth Verification

The ground truth carries one row per claim either instrument can reach: 164 rows against the 178 numeric tokens extracted from the article and appendix. The claims left out are scale endpoints and category labels the deposit cannot reach, and numbers the article takes from other papers; they are recorded in `published_claims.csv` and verified against the sources they come from.

Table 3: Ground truth verification by component (164 rows)

Table or figure	Not evaluable	Does not match	Matches
Analysis, footnote 5	2	0	0
Descriptive	7	2	32
Design	0	0	16
Discussion	0	0	1
Figure 3	0	2	2
Figures 3 to 8, B.1, B.2, C.3, C.4	0	0	1
Figures 5 and 6	1	0	0
Observational text	2	4	10
Table 1	0	0	30
Table 2	0	0	4
Table A.1	40	0	8

Every row that does not match, or that could not be evaluated for a reason specific to it, is below. The 40 Table A.1 cells the deposit cannot reach share one reason between them and are counted in the table above rather than listed one line each.

Table 4: Every claim that does not reproduce, and every one that cannot be evaluated

Claim	Published	Pipeline	Locus
Distinct municipalities	225	237	paper_internal
Partisan elections, more experienced candidate wins (%)	62	61.29	paper_internal
Nonpartisan elections, more experienced candidate wins (%)	61	62.06	paper_internal
p-value on the partisan/nonpartisan difference	0.791	0.8516	paper_internal
Political experience, MTurk nonpartisan, smallest effect (pp)	25	22.65	paper_internal
Political experience, MTurk nonpartisan, largest effect (pp)	30	31.97	paper_internal

Democrat preference for a female over a male candidate (pp)	10	–	unresolved
Partisan elections, attorneys or business executives (%)	63	–	archive
Nonpartisan elections, attorneys or business executives (%)	48	–	archive
Agreement between the OLS and cjoint AMCE estimates	–	–	archive
Agreement between the OLS and logit marginal effects	–	–	archive
The MTurk sample is whiter, more male, more liberal, more Democratic, better educated and younger	–	–	archive
Figure 3: the job experience, race, age and gender effects are relatively muted	–	–	–
Figure 4: the gender and age effects are as small as in the MTurk sample	–	–	–
Figures 5 and 6: the nonwhite effects are positive for Democrats and negative for Republicans	–	–	paper_internal
Figure 5: the race and gender patterns do not differ much by election type	–	–	–
Figure 7: the competence differences clear zero for Democrats and not for Republicans	–	–	paper_internal
Figures 7 and 8: the race, gender and age effects on competence are small	–	–	–
The candidate types rated most competent are the ones preferred in nonpartisan elections	–	–	–
Appendix C: the competence results resemble the candidate preference results	–	–	–

Summary: 104 of 112 evaluable claims match the published values; 8 do not. `ground_truth/build_ground_truth.R` rebuilds this table from `maintained/output/` at the end of every run. The only numbers typed into that script are the published values themselves, which serve as comparison targets and are never inputs to a computation.

The archive columns are empty, and that is a property of the deposit. `value_script` and `match` are `NA` on every row. The deposited scripts write no files at all, keeping every result as an in-memory R object, as the archive's own README explains; nothing in this repository re-derives those objects, so there is no archive number to read back. The comparison that can be made is between the article and the maintained rewrite, which is what `value_rewrite` and `match_rewrite` record. The one place the archive's own output has been compared directly is the 729 figure estimates described under Figure Verification.

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6 Maintained Rewrite

The rewrite lives in `maintained/`: a shared `helpers.R`, one script per published table or figure, four for in-text quantities, and the claims file that prints every published number beside the sentence that states it.

Table 5: Maintained rewrite scripts

Script	Output
helpers.R	Packages, both datasets, the estimation and plotting helpers
table_1_observational_experience.R	output/table_1_observational_experience.csv, output/text_observational_chisq.csv, output/text_observational_dataset_description.csv
table_a1_party_identification.R	output/table_a1_party_identification.csv
figure_3_mturk_main_candidate_pref.R	output/figure_3_mturk_main_candidate_pref.{pdf,png,csv}
figure_4_yougov_main_candidate_pref.R	output/figure_4_yougov_main_candidate_pref.{pdf,png,csv}
figure_5_mturk_het_fx_candidate_pref.R	output/figure_5_mturk_het_fx_candidate_pref.{pdf,png,csv}
figure_6_yougov_het_fx_candidate_pref.R	output/figure_6_yougov_het_fx_candidate_pref.{pdf,png,csv}
figure_7_mturk_het_fx_competence.R	output/figure_7_mturk_het_fx_competence.{pdf,png,csv}
figure_8_yougov_het_fx_competence.R	output/figure_8_yougov_het_fx_competence.{pdf,png,csv}
figure_b1_mturk_satisficing.R	output/figure_b1_mturk_satisficing.{pdf,png,csv}
figure_b2_yougov_satisficing.R	output/figure_b2_yougov_satisficing.{pdf,png,csv}
figure_c3_mturk_main_competence.R	output/figure_c3_mturk_main_competence.{pdf,png,csv}
figure_c4_yougov_main_competence.R	output/figure_c4_yougov_main_competence.{pdf,png,csv}
text_design_parameters.R	output/text_design_parameters.csv
text_observational_experience_advantage.R	output/text_observational_experience_advantage.csv
text_mturk_yougov_correlation.R	output/text_mturk_yougov_correlation.{csv,pdf,png}
text_policy_index_het_fx.R	output/text_policy_index_het_fx.csv
in_text_claims.R	Printed to the console: every published number beside its sentence

Key substitutions:

Table 6: Deprecated patterns and replacements

Original	Rewrite
rm(list = ls())	removed
library() in each script	source(helpers.R); all packages in helpers.R
custom cl() over sandwich and lmtest	lm_robust(clusters = , se_type = 'stata'), identical to 1e-15
coefficient rows selected by index	filter(str_detect(term, ':nonpartisan$'))
labels reassigned from another model's rownames	labels derived from each model's own terms
position_dodgev() from ggstance via coefplot	position_dodge(width = 0.5)
geom_errorbarh(height = 0)	geom_linerange()
geom_segment() for confidence bars	geom_linerange()
gather / spread	pivot_longer / pivot_wider
magrittr pipe	native pipe
figures held only as in-memory R objects	written to output/ as PDF, PNG and CSV
load('yougov_replication.rdata')	load('yougov_replication.xlsx'), the deposited name

The standard error substitution. The archive's `cl()` function scales the CR0 meat by `(M / (M - 1)) * ((N - 1) / (N - K))`, which is what `estimatr::lm_robust(se_type = "stata")` computes. The two agree to 1e-15 on every model in the paper. An earlier version of this rewrite asked for `se_type = "CR1S"`, the name estimatr used for the same estimator until version 1.0.6 retired it; that name now raises an error, which is why every figure script here names the estimator "stata".

The weighting decisions are surfaced, not defaulted. The archive applies the YouGov survey weights to the main-effects figures (4 and C.4) and to the satisficing figure (B.2), but

not to the heterogeneous-effects figures (6 and 8) and not to the four models behind the correlation in the Discussion. The rewrite reproduces each figure as the archive specifies it, and `election_type_panels()` takes `weighted` as a required argument with no default so the choice is visible at every call site. Weighting the heterogeneous-effects models would move individual estimates by up to 0.04 and the reported correlation from 0.95 to 0.92, which is a change to the analysis rather than a repair to it.

What the rewrite adds. The published appendix figures B.1 and B.2, which the archive computes and no earlier rewrite covered, and the one panel of Table A.1 the deposit can support. The in-text observational comparison on page 5 has no code anywhere in the archive or the recovered files; it is reconstructed here from the recovered data. `text_design_parameters.R` reads back the design facts the article states in prose (elections per subject, attributes per election type, the age levels, the endpoints of the competence scale) rather than leaving them asserted. Every figure now writes a CSV of the plotted estimates alongside the image, which is what makes a rerun diffable.

The policy-goal analysis is rewritten even though it was never published. The Mechanism section reports results for a policy-goal index and cites "Appendix C.2" for them. There is no Appendix C.2: the published appendix has sections A, B and C only, C has no subsections, and the policy-goal analysis appears nowhere in it. Both deposited scripts compute it, so `text_policy_index_het_fx.R` reproduces those models and the two sentences the article bases on them can be checked. The index's direction is documented nowhere, and the appendix section that would have defined it does not exist, so the candidate's own party label is used as the anchor: a Republican candidate raises the index and a Democratic candidate lowers it in every sample by respondent party cell, which fixes the high end as the conservative end. Read that way both sentences hold.

What the rewrite leaves alone. The confidence bars are drawn at plus or minus 1.96 standard errors, as the archive draws them, rather than at a t quantile on the cluster-adjusted degrees of freedom. The valence-index analysis the archive computes is not rewritten, since no sentence in the article or the appendix refers to it. The unweighted YouGov specifications stay unweighted.

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7 Figure Verification

Every plotted coefficient and standard error in the ten conjoint figures was compared against the archive's own `lm()` plus `cl()` output, term by term and panel by panel: 729 pairs, agreeing to within 4e-13. Two of the figures follow.

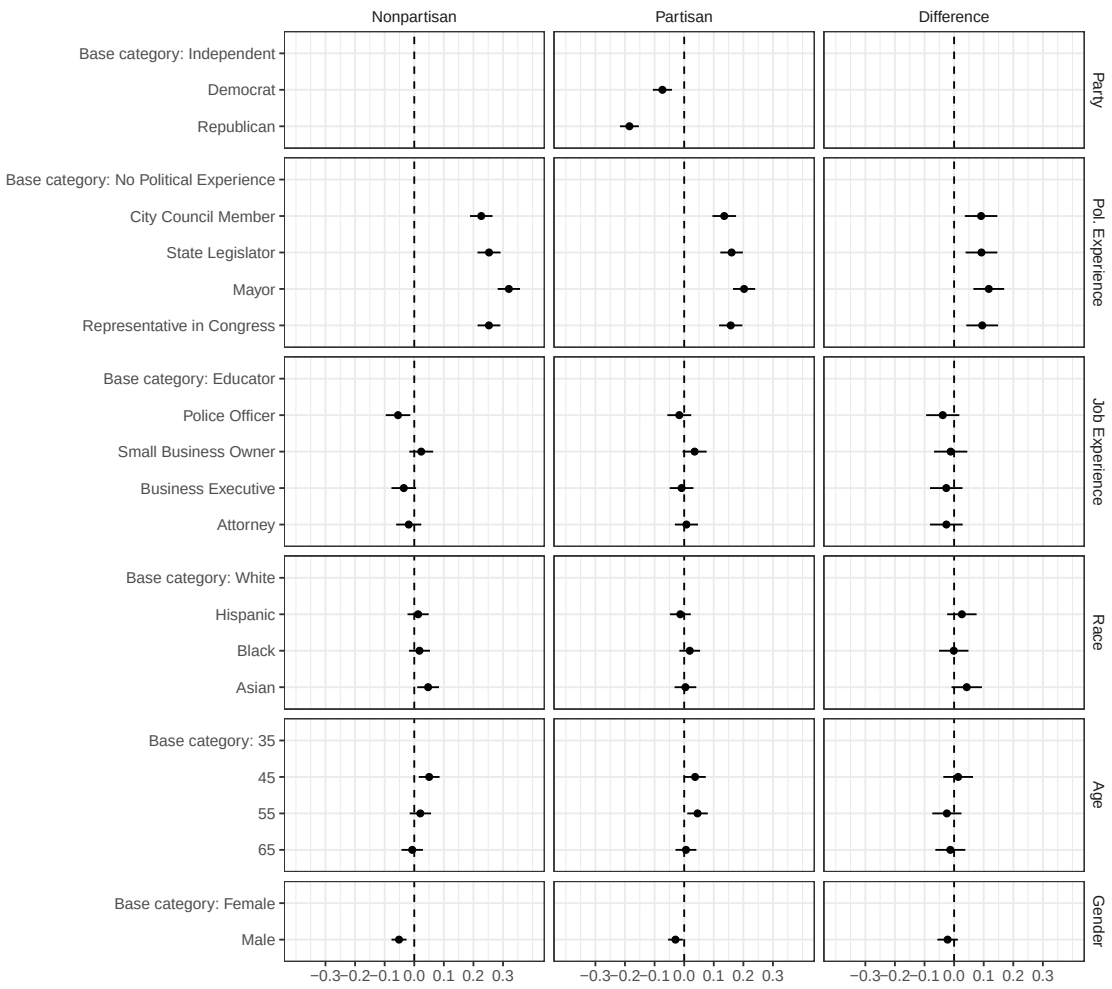


Figure 1: Figure 3 reproduction. MTurk main analysis, candidate preference. Columns are nonpartisan elections, partisan elections, and the difference between them; rows are attribute groups. Political experience carries visibly more weight when the party label is withheld.

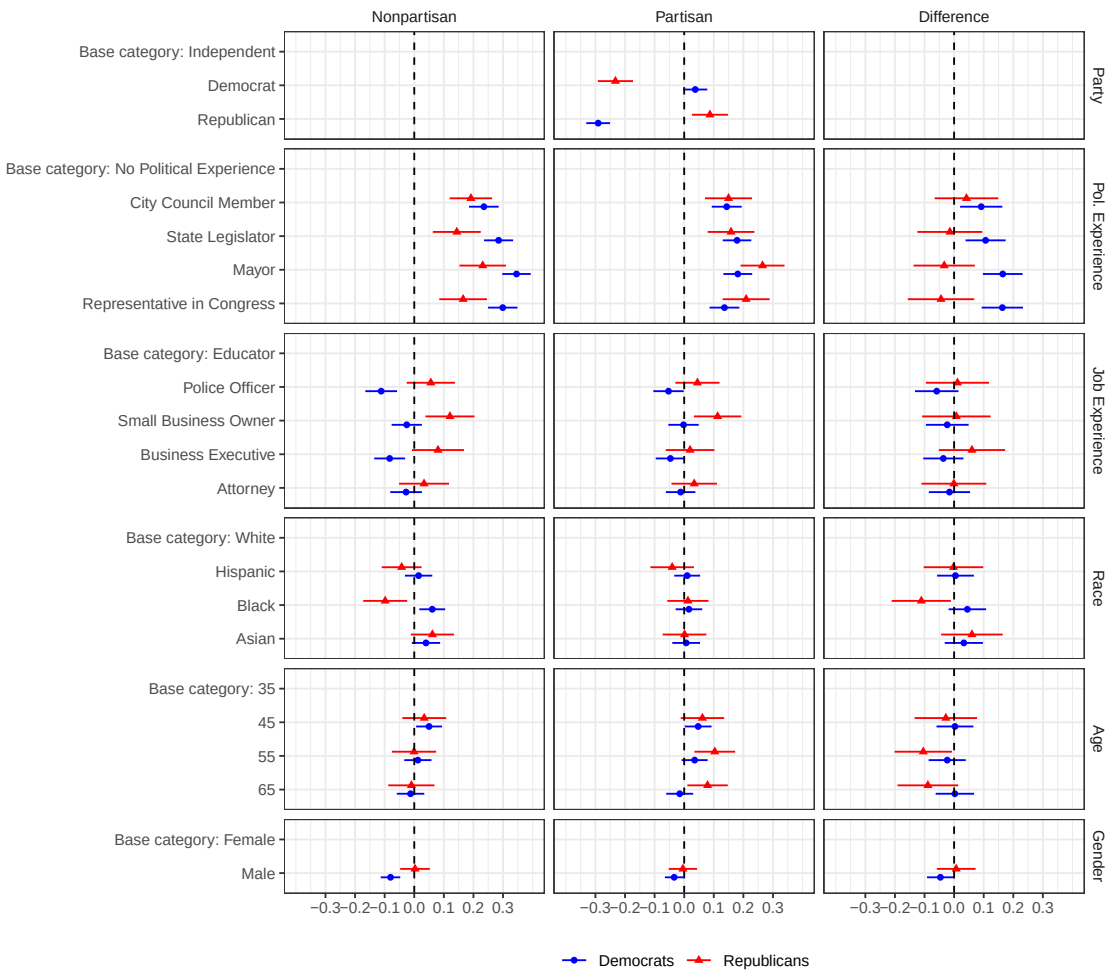


Figure 2: Figure 5 reproduction. MTurk heterogeneous effects on candidate preference, Democratic respondents in blue and Republican respondents in red.

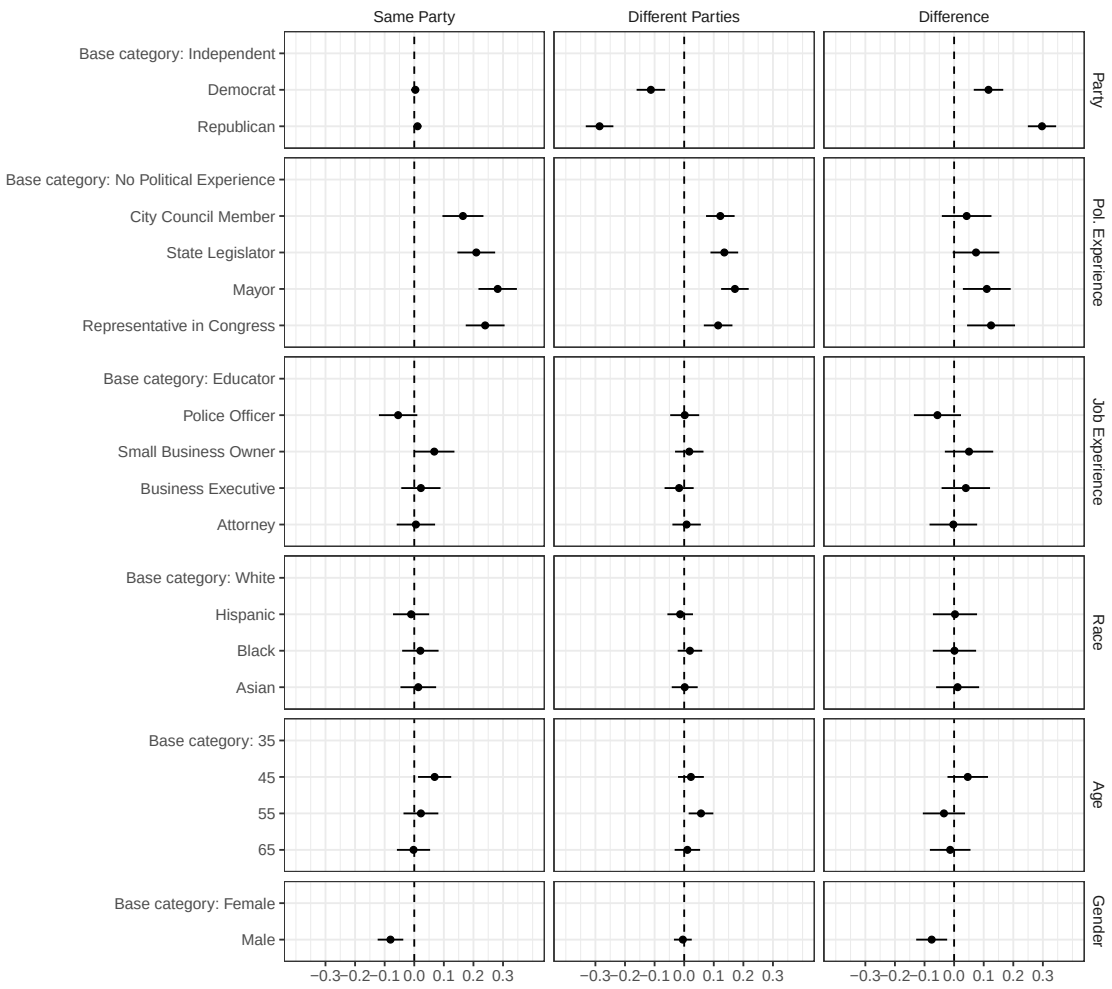


Figure 3: Appendix Figure B.1 reproduction, the satisficing check. Partisan elections only, split by whether the two candidates share a party. Both panels display six attributes, so the number of attributes is held fixed and only the informativeness of the party label varies.

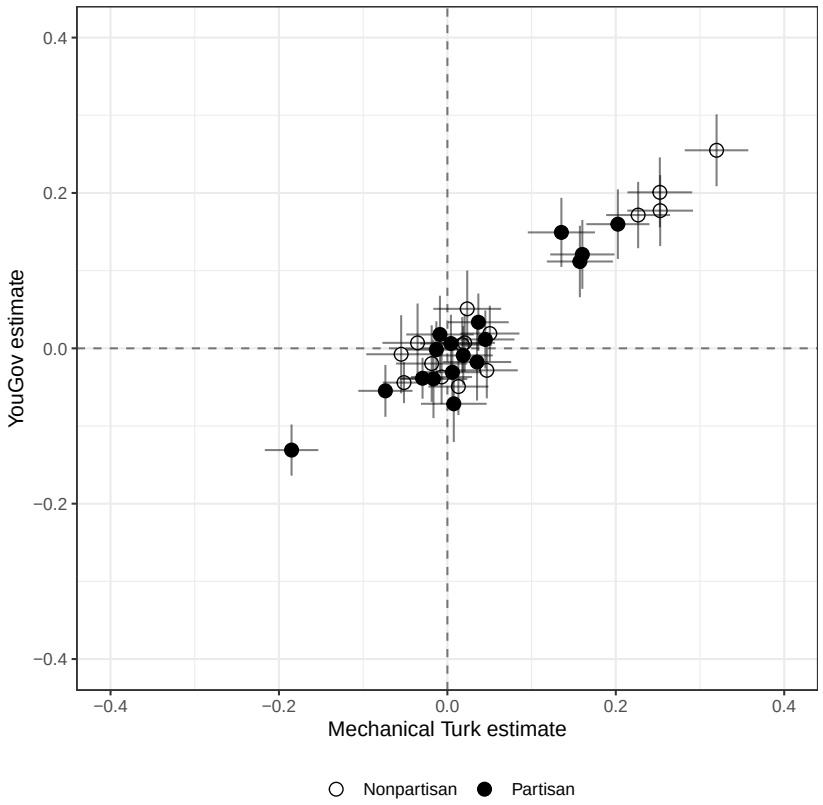


Figure 4: The Discussion's correlation of 0.95, plotted. Each point is one attribute level's effect on candidate preference, estimated on MTurk and on YouGov, with 95 per cent intervals on both axes.

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8 In-Text Claims

`maintained/in_text_claims.R` carries one block per published claim: the article's sentence verbatim in a comment, then code that reads `maintained/output/` and prints the number in the article's own units and rounding. It never fits a model and never reads the ground truth. It reaches the same quantities as `build_ground_truth.R` by its own route through the same output files, so a disagreement between the two is a finding rather than a coincidence.

Descriptive claims get a computed truth value rather than a number, and the evidence behind it is printed on the line beneath. A claim that reduces to a boolean gets a verdict; "relatively muted", "similarly small" and "do not differ much" set no threshold, so those blocks print

the distribution and no verdict, because a verdict would invent the threshold the sentence withholds.

Coverage is a gate rather than a convention. `build_ground_truth.R` runs the claims file non-interactively, reads the claim ids back off the lines it actually printed, and halts unless every pipeline and descriptive row of `published_claims.csv` appears among them, unless every id printed is one the article makes, and unless the ids declared in the blocks and the ids printed at run time are the same set. Matching the declarations as text would pass a block that errors or prints nothing.

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9 Maintained Rewrite Verification

Two full runs of `run_all.R` produce byte-identical CSV and PNG output. Only the PDF figures differ between runs, because a PDF records the time it was written.

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10 R Environment

Table 7: Key package versions

Package	Version
tidyverse	2.0.0
estimatr	1.0.6
knitr	1.51
kableExtra	1.4.0
here	1.0.2

R version: R version 4.6.0 (2026-04-24)